위상 양자 물질 이론 연구실

황규성 교수님



KYUNG HEE UNIVERSITY
Dept. of Applied Physics



Topological Quantum Matter
Theory Group

Principal Investigator



Kyusung Hwang (황규성)

Associate Professor

Department of Applied Physics
Kyung Hee University (Global Campus)
Office: Rm 141-2, Building of College of Applied Sciences
e-mail: kyusung.hwang@khu.ac.kr
Google Scholar ORCID

Employment

- Mar. 2025~present: **Associate Professor, Kyung Hee University**
- Apr. 2022~Feb. 2025: **Assistant Professor(KIAS Fellow), Korea Institute for Advanced Study**
- Sept. 2018~Mar. 2022: **Postdoctoral Research Fellow, Korea Institute for Advanced Study**
- Oct. 2016~Jul. 2018: **Postdoctoral Research Fellow, The Ohio State University**

Education

- Jan. 2012~Sep. 2016: **Ph.D. in Physics, University of Toronto**
  "Novel Quantum Phases in Correlated Electron Systems" (advisor: Prof. Yong Baek Kim)
- Sep. 2008~Aug. 2010: **M.S. in Physics, Seoul National University**
- Mar. 2001~Feb. 2008: **B.S. in Physics, Yonsei University** (military service: Jun. 2003~Nov. 2005)

Group Members



Jeong Hwan Hwang (황정환)

Graduate student (Master program)

Research topics: Topological phases, Quantum spin liquids

e-mail: curlumr@khu.ac.kr



Yong Chan Kim (김용찬)

Undergraduate student

Research topics: Topological magnons, Quantum geometry, Kitaev magnetism

e-mail: yongchan1@khu.ac.kr



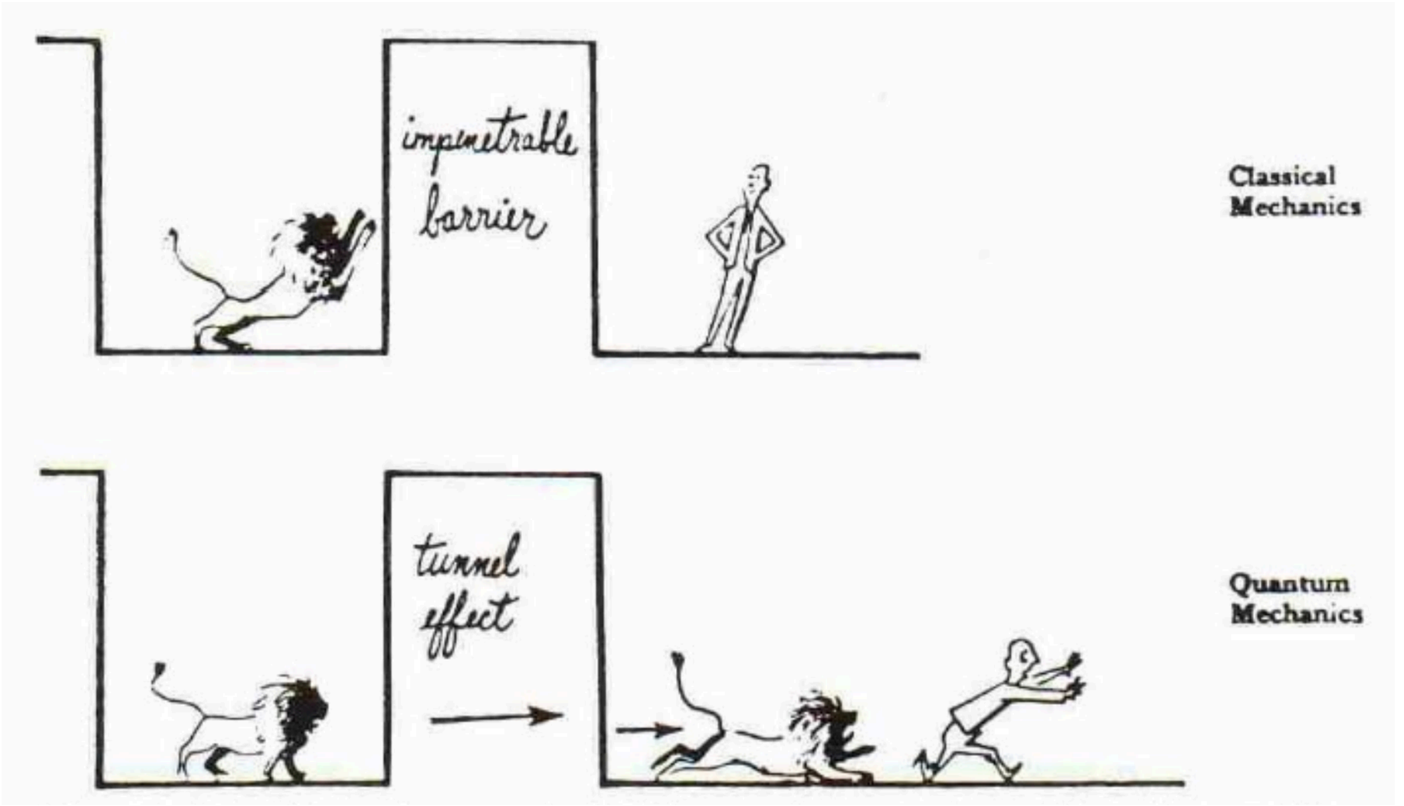
Seongwoo Choi (최성우)

Undergraduate student

Research topics: Topological phases

e-mail: cws0118@khu.ac.kr

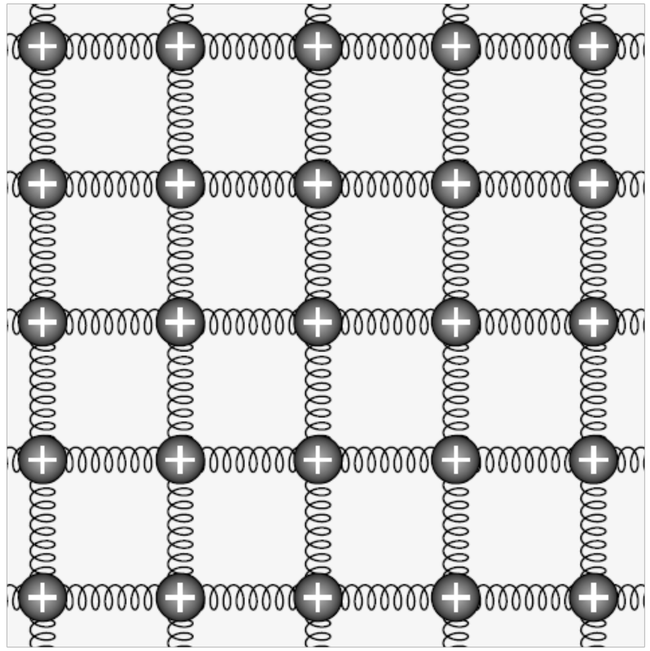
양자 물질 ?

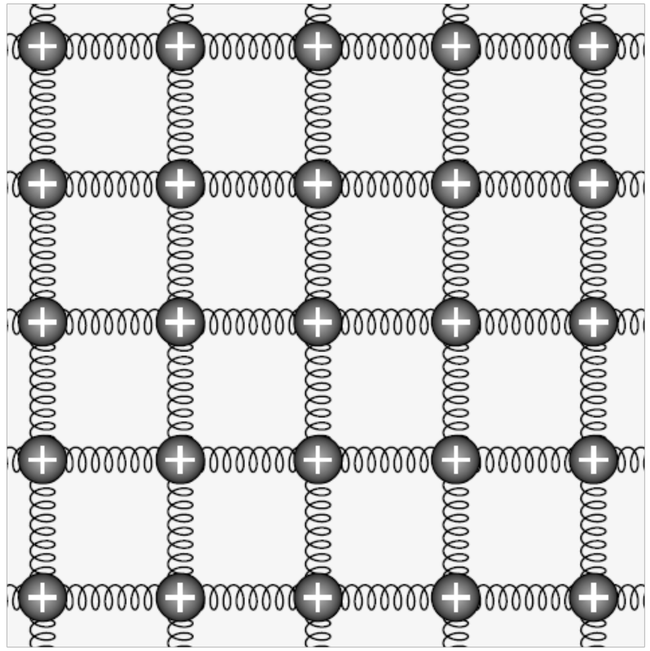


Van Vleck (1979)

Adobe Stock | #970940563



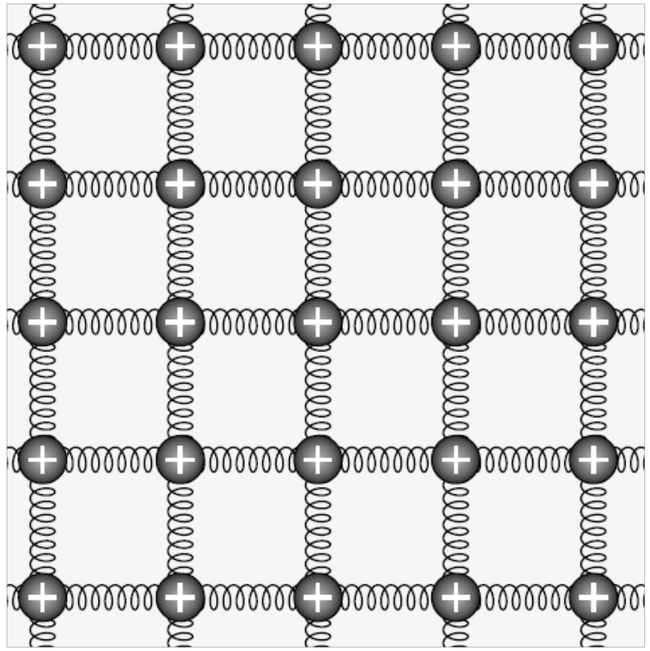




고전역학

$$E(N) \lesssim c' N^{5/3} \implies \frac{E(N)}{N} \sim N^{2/3} \to \infty$$

Dyson–Lenard (1967)



고전역학

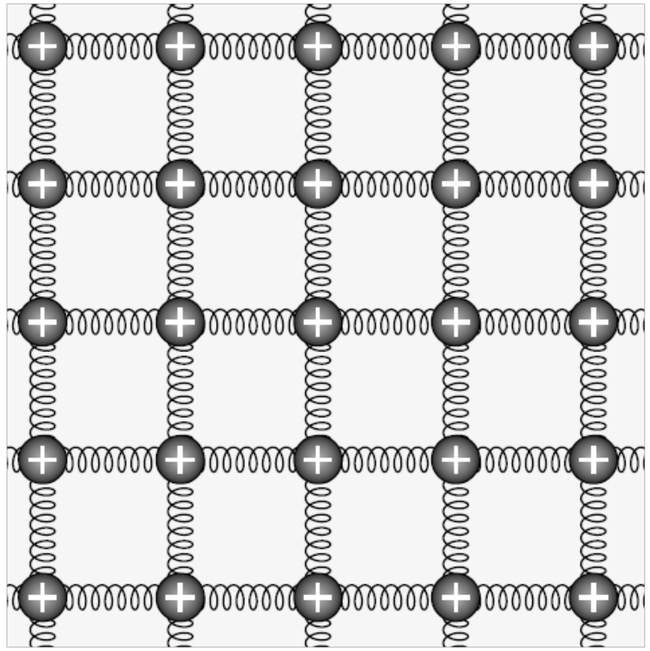
$$E(N) \lesssim c' N^{5/3} \implies \frac{E(N)}{N} \sim N^{2/3} \to \infty$$

Dyson–Lenard (1967)

양자역학
Pauli 배타원리

$$E(N) \lesssim c'' N \implies \frac{E(N)}{N} \sim \text{const.}$$

Dyson–Lenard (1968)



고전역학

$$E(N) \lesssim c' N^{5/3} \implies \frac{E(N)}{N} \sim N^{2/3} \to \infty$$

Dyson–Lenard (1967)

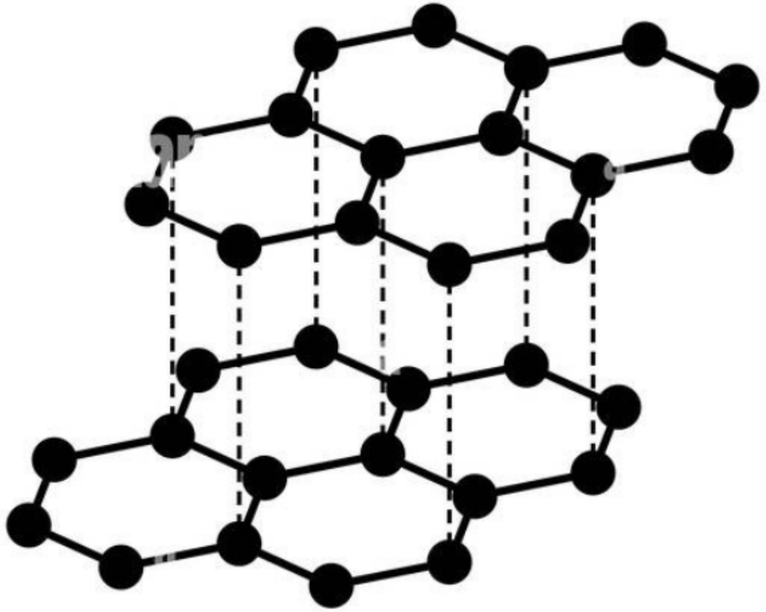
응집물질은 양자역학 없이 설명될 수 없다

양자역학
Pauli 배타원리

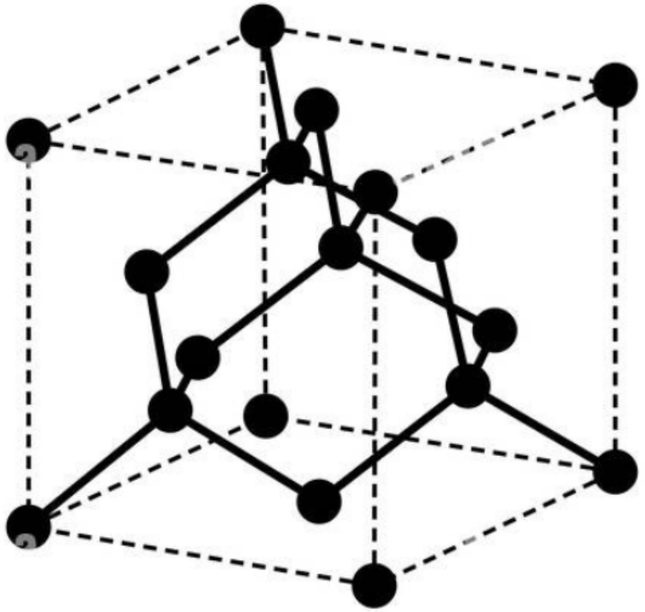
$$E(N) \lesssim c'' N \implies \frac{E(N)}{N} \sim \text{const.}$$

Dyson–Lenard (1968)

흑연



다이아몬드



전자의 배치, 구조에 따라서 물질의 성질이 달라진다.

양자 물질

양자 물질 : 양자역학적 효과가 **거시적인 스케일에서** 나타나는 물질

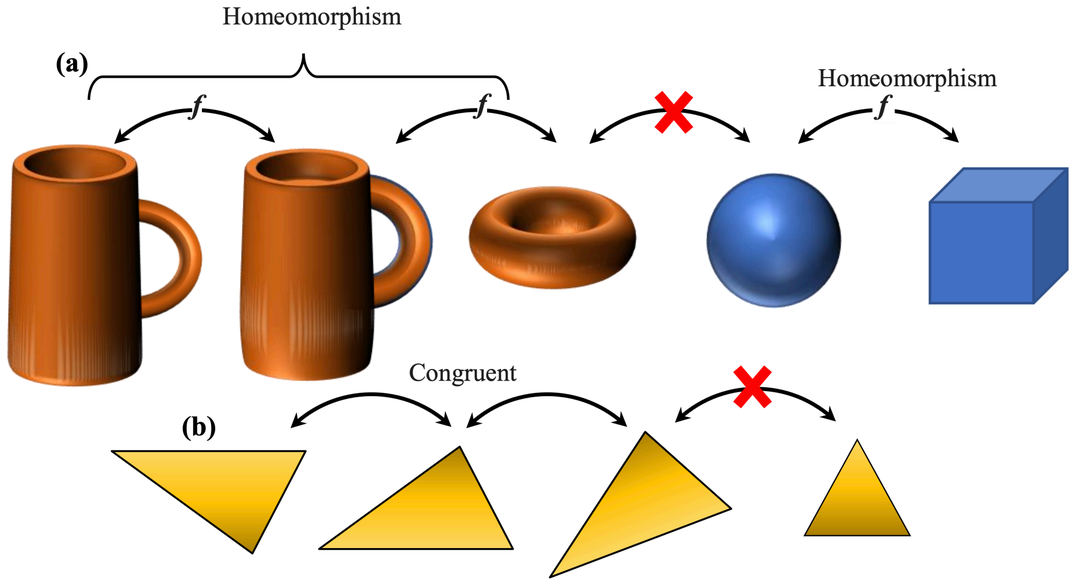
- **초전도체**
- **그래핀**
- **자성체**
- **양자 스핀 액체**
- **위상 절연체**
- **Weyl 준금속**

양자 물질

양자 물질 : 양자역학적 효과가 **<u>거시적인 스케일에서</u>** 나타나는 물질

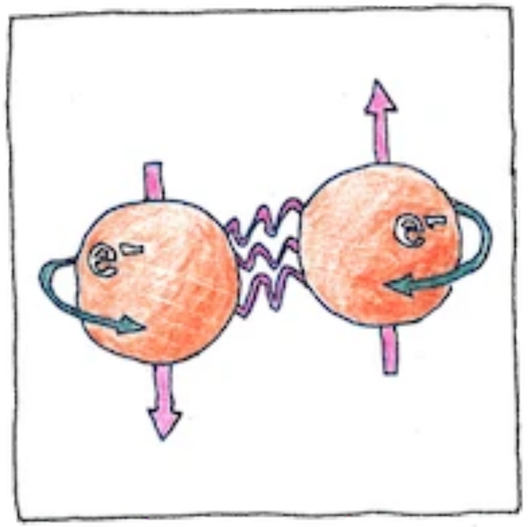
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- 그래핀
- 자성체
- 양자 스핀 액체
- 위상 절연체
- Weyl 준금속

위상 양자 물질



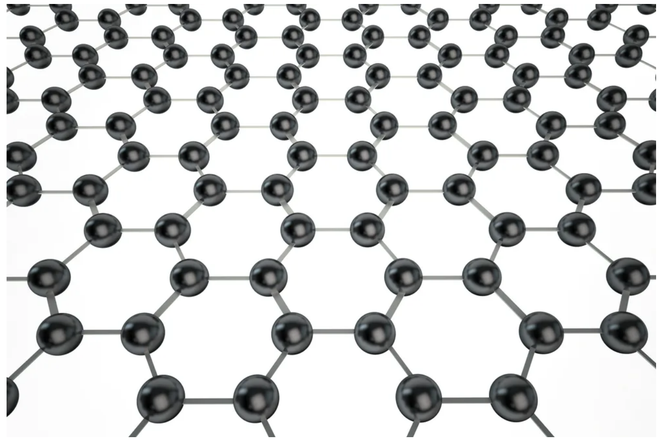
양자 물질

• 초전도체



전자들이 짝을 이뤄 거시적 양자 상태로 **응축하여**, **저항 없이** 전류가 흐르는 물질

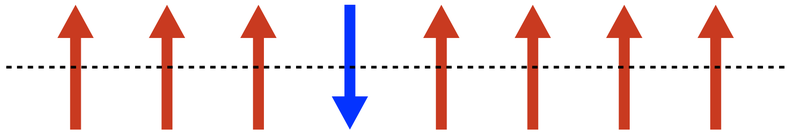
• 그래핀



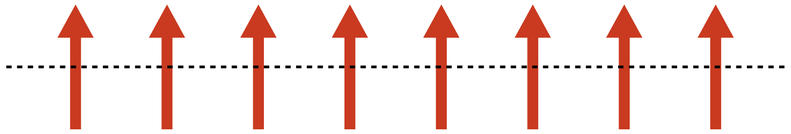
탄소 원자의 벌집 격자에서 전자가 질량이 없는 **Dirac 페르미온**처럼 거동함

양자 물질

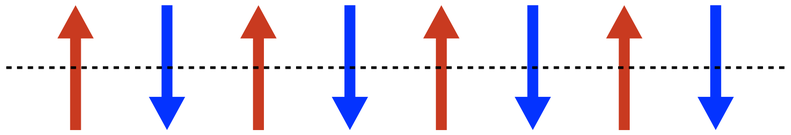
- 자성체



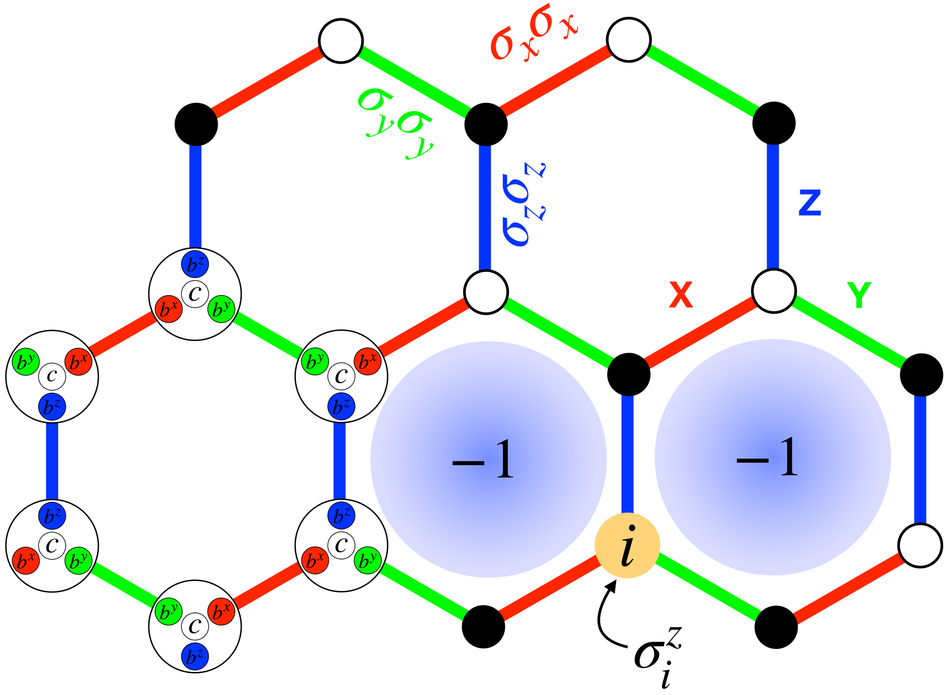
single spin flip = 1 magnon (quasi-particle)



강자성체

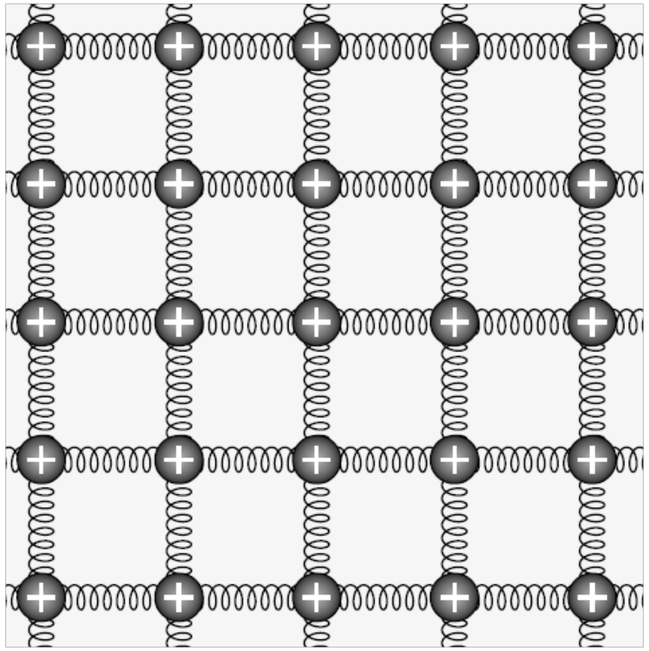


반강자성체

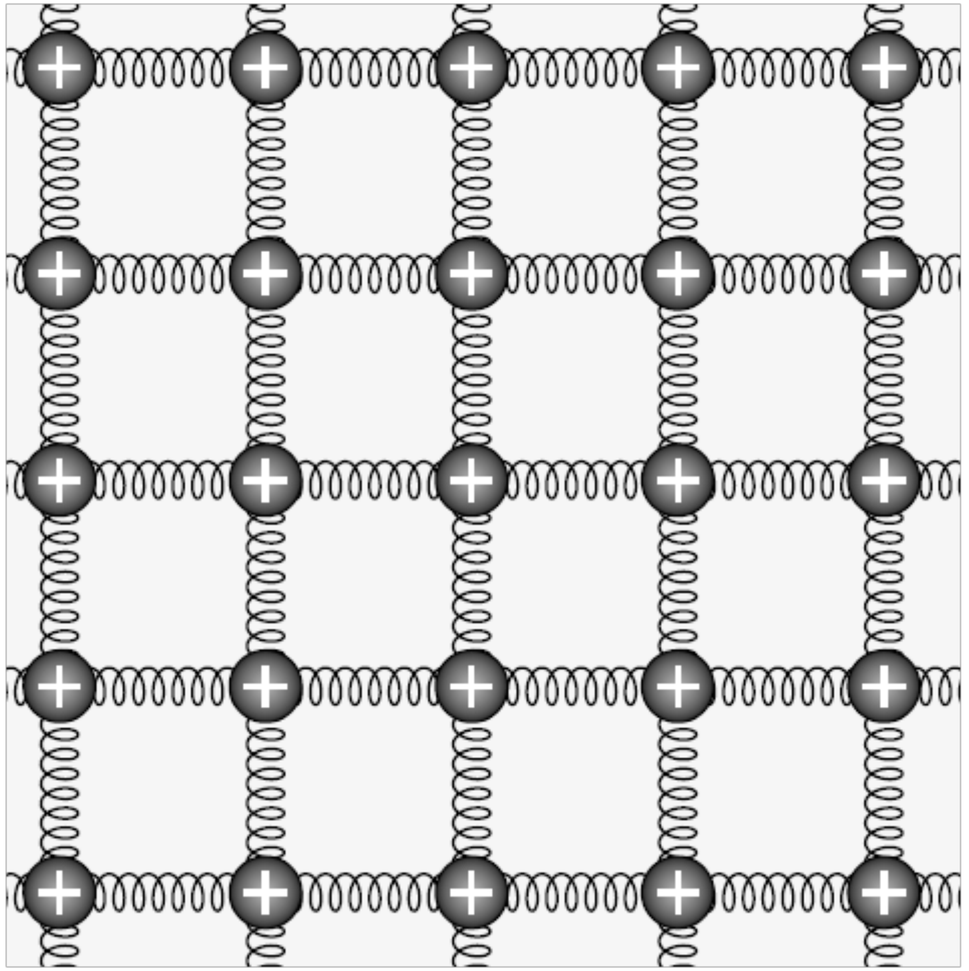


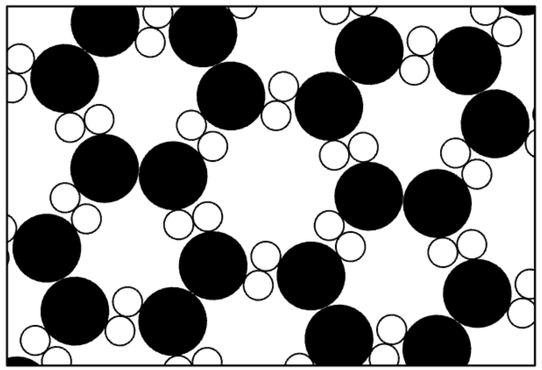
양자 스핀 액체

위상 ?

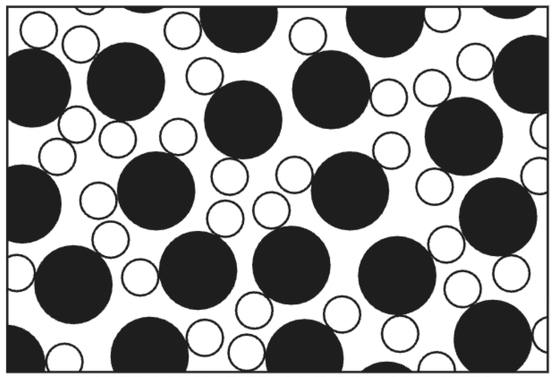


대칭성?

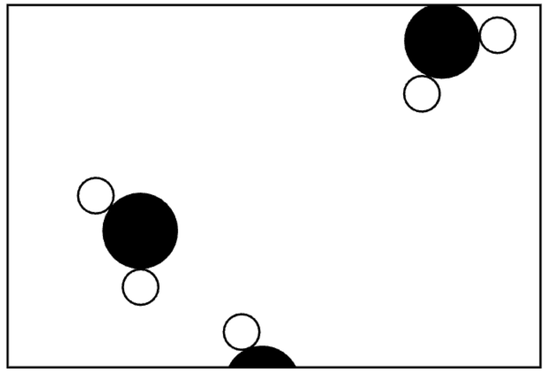




ICE



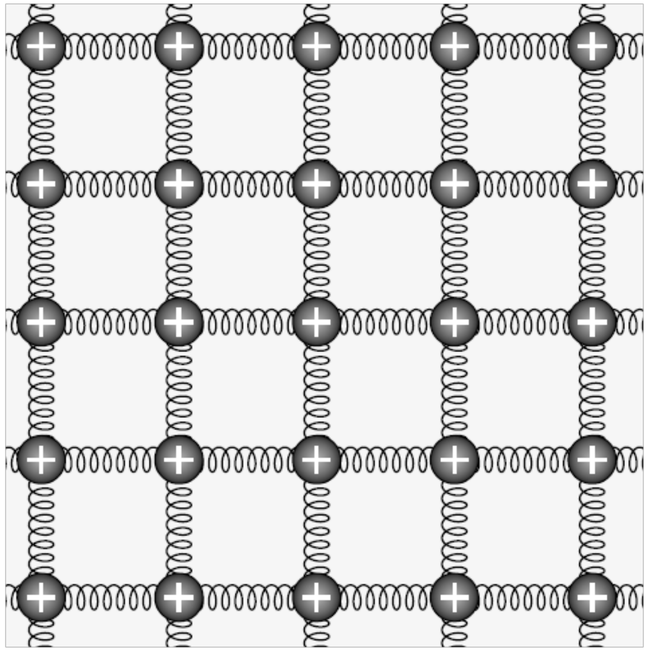
WATER MAGNIFIED ONE BILLION TIMES

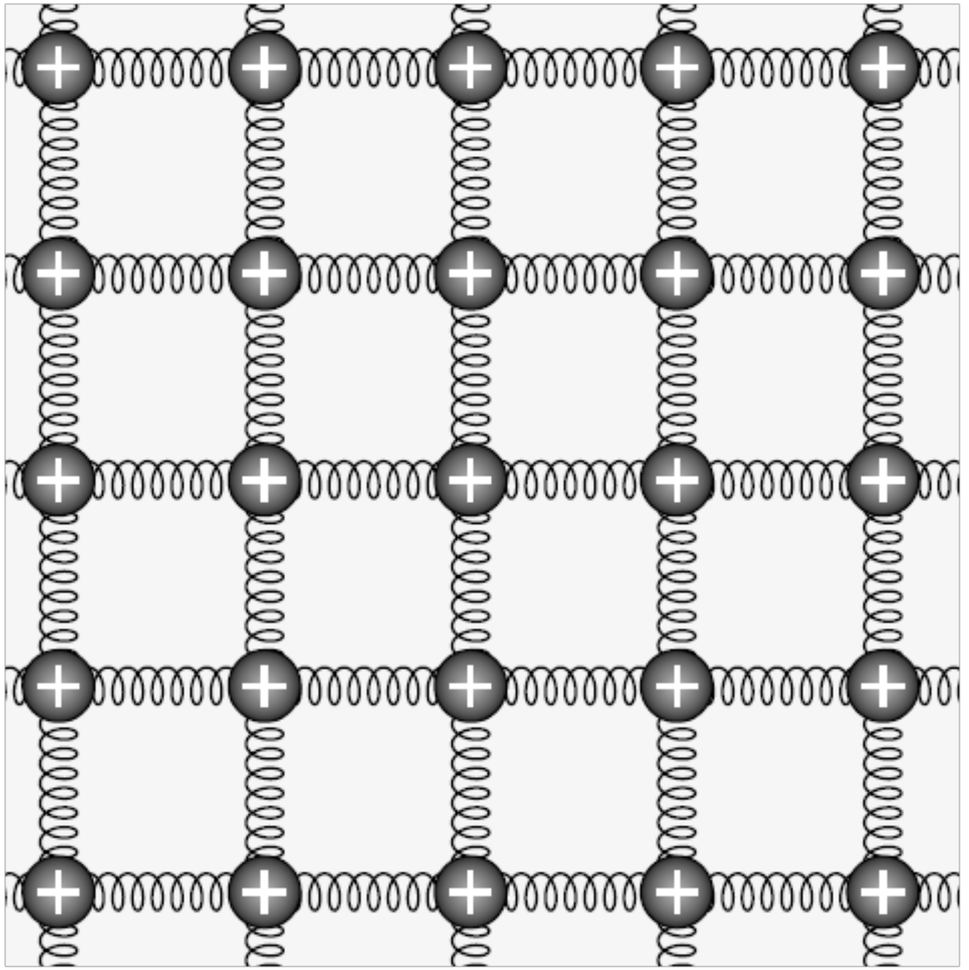


STEAM

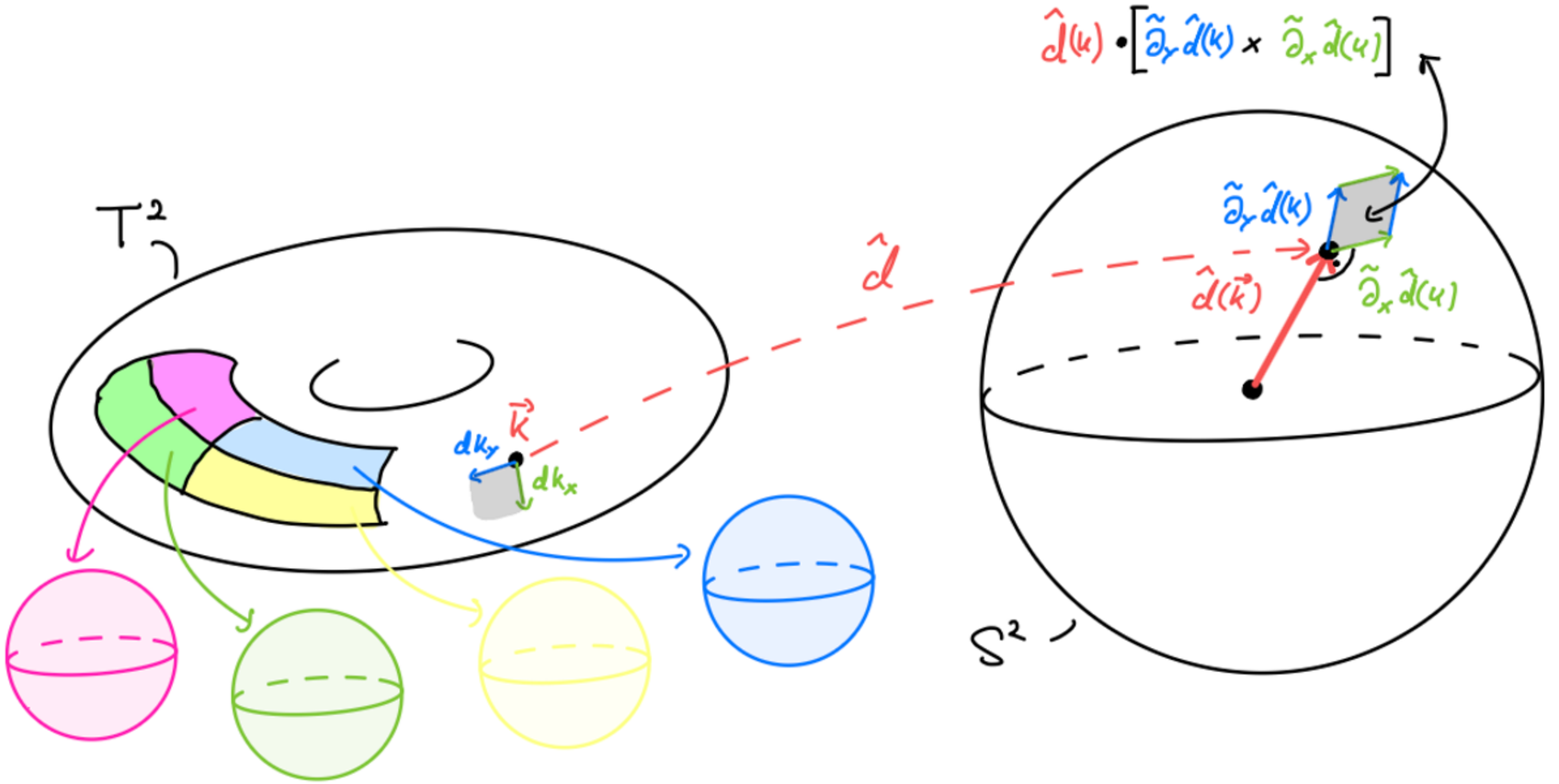


대칭성





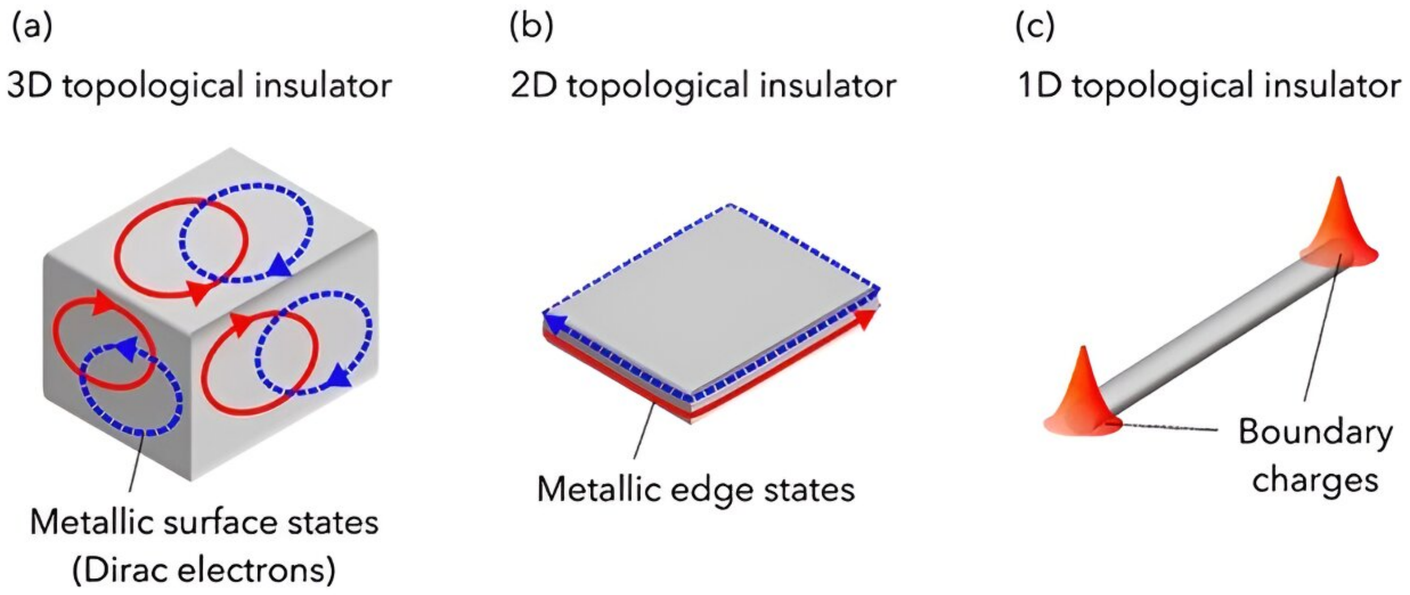
대칭성을 공유하더라도 서로 다른 상인 물질들을 분류할 필요가 있다 → **위상 불변량**



대칭성을 공유하더라도 서로 다른 상인 물질들을 분류할 필요가 있다 → **위상 불변량**

양자 물질

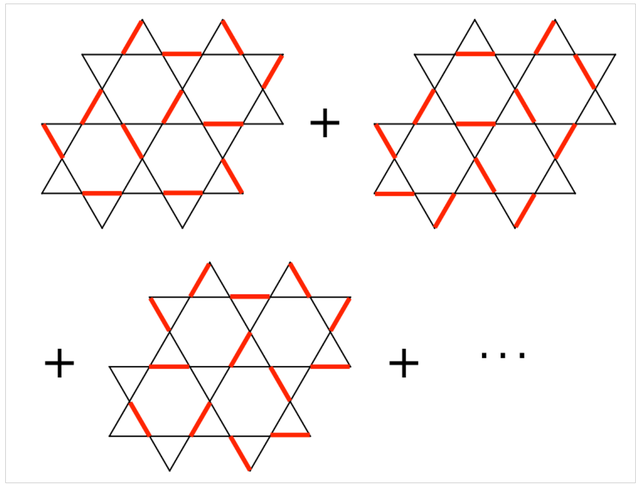
- **위상 절연체**



Bulk는 절연체지만 경계에 위상학적 도체 상태(**edge mode**) 가 존재하는 물질

위상 양자 물질

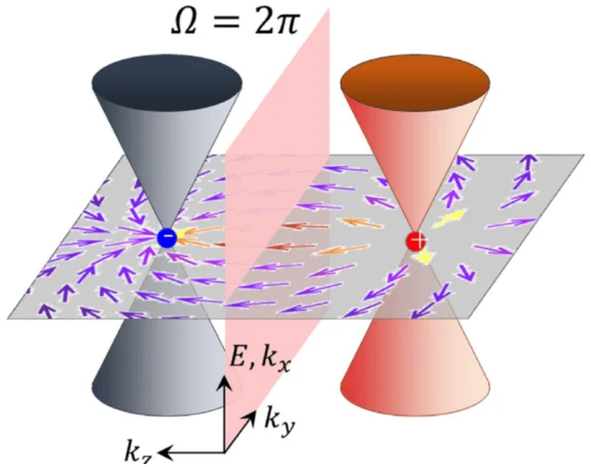
• 양자 스핀 액체



Resonating Valence Bond (RVB) state
by P. Anderson

절대 영도 까지도 **자기 정렬을 이루지 않고**,
거시적으로 **얽혀있는** 물질

• Weyl 준금속

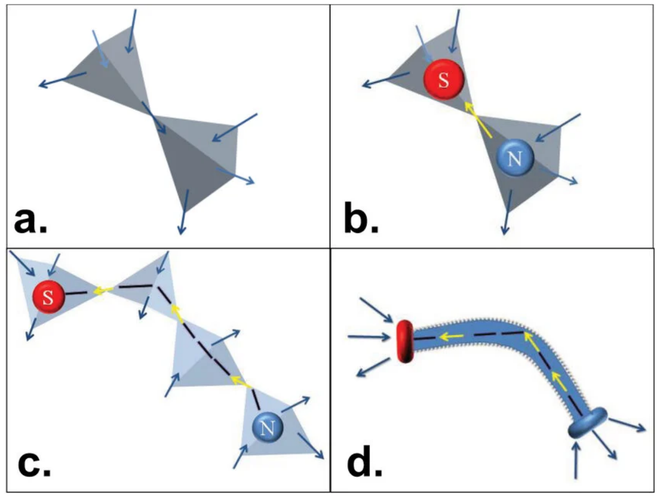


운동량 공간에 **chirality** ± 1**을 가진 점이**
짝으로 등장하는 3D 물질

왜 다루는가 ?

고에너지 물리학에서 가설로 존재하던 입자들이 응집물질물리학에서도 존재한다.

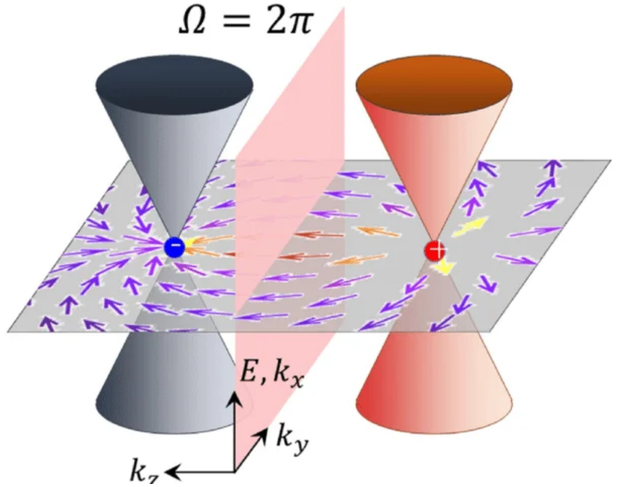
실공간



자기 단극자

Dirac (1931)

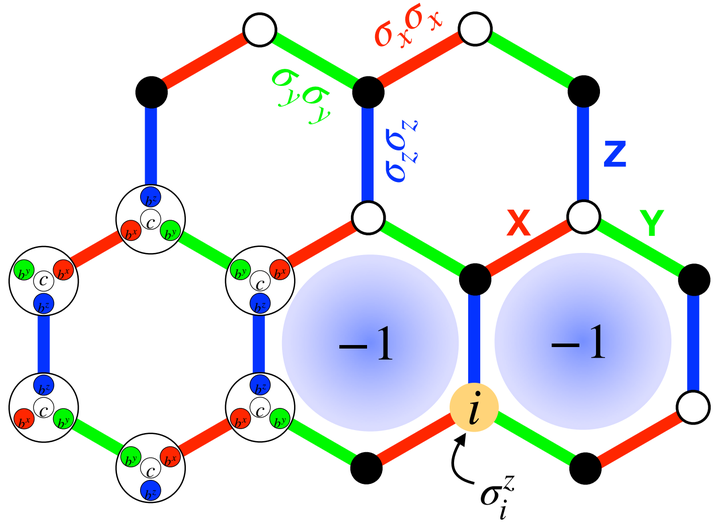
운동량 공간



Weyl 점 / Berry 단극자

Wely (1929)

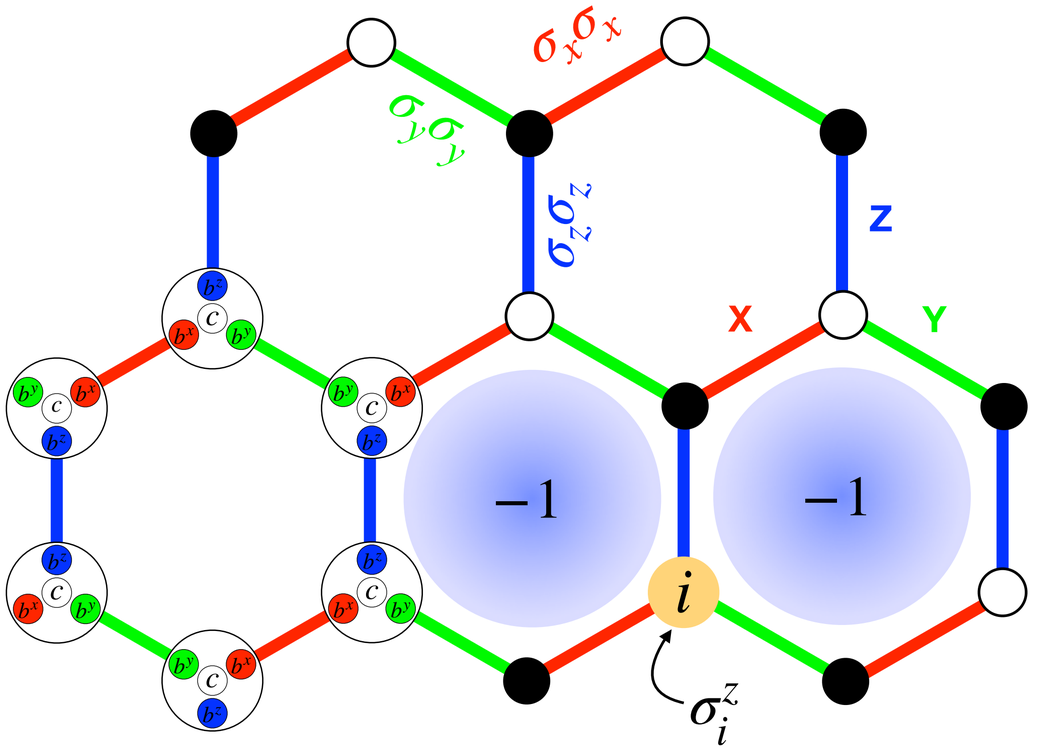
분수화



Majorana 페르미온

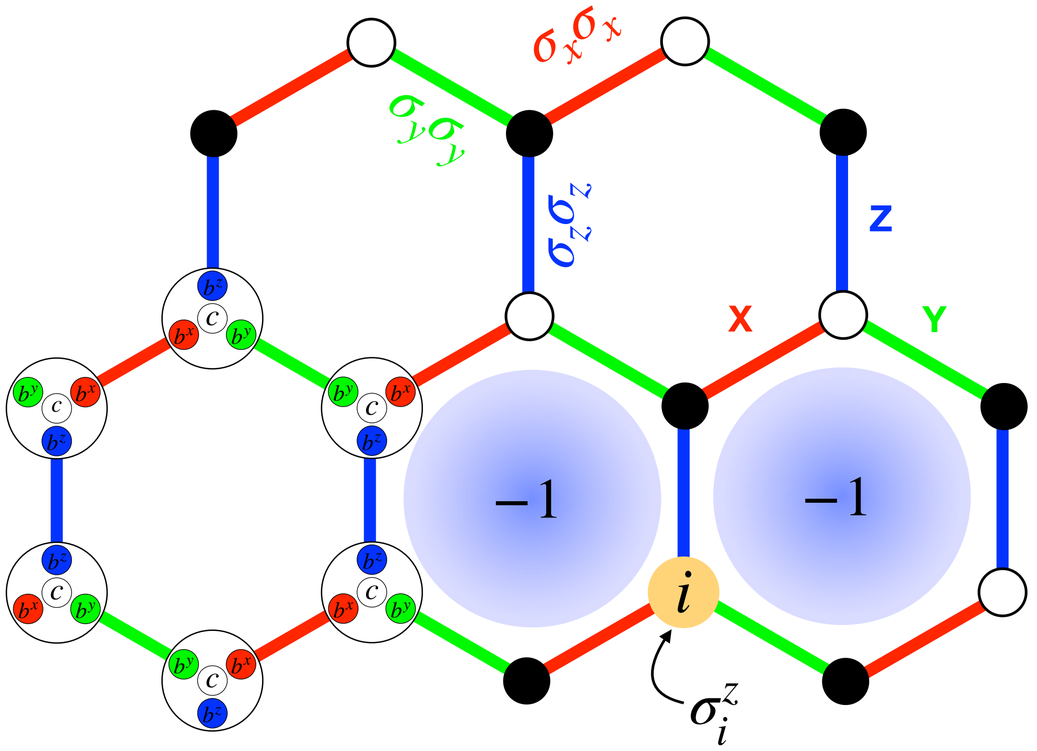
Majorana (1937)

Kitaev 양자 스핀 액체



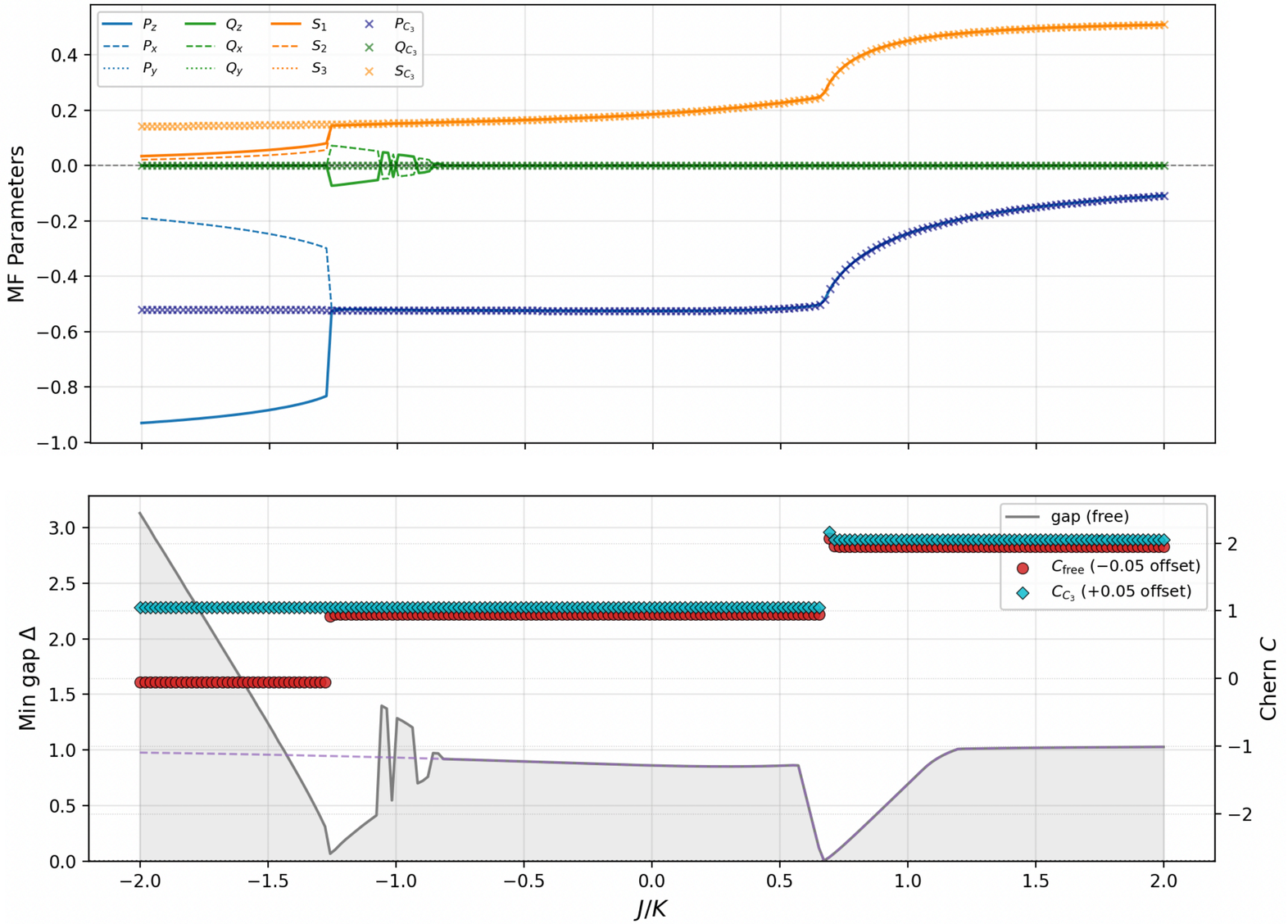
$$H = \underbrace{J\sum_{\langle ij\rangle_x}\sigma_i^x\sigma_j^x + J\sum_{\langle ij\rangle_y}\sigma_i^y\sigma_j^y + J\sum_{\langle ij\rangle_z}\sigma_i^z\sigma_j^z}_{\text{Pure Kitaev model}} + \underbrace{J_Q\sum_{\langle ij\rangle_z,\,\langle \ell m\rangle_z}\left(\sigma_i^x\sigma_j^x\right)\left(\sigma_\ell^y\sigma_m^y\right)}_{\text{Inter-bond } c^4 \text{ interaction}}$$

Kitaev 양자 스핀 액체



$$H = \underbrace{-iJ \sum_{\alpha=x,y,z} \sum_{\langle ij \rangle_\alpha} u_{ij}^{\alpha}\, c_i c_j}_{\text{Free Majorana hopping}} \underbrace{- J_Q \sum_{\langle ij \rangle_z,\, \langle \ell m \rangle_z} u_{ij}^{x}\, u_{\ell m}^{y}\, (ic_i c_j)(ic_\ell c_m)}_{c^4 \text{ inter-bond interaction}}$$

Kitaev 양자 스핀 액체




SUMMIT
400m to go

감사합니다 !